

## ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

## Chapter 7: Inertial Reference Systems & Ring Laser Gyros

Syllabus Source: CAE Oxford (pp. 229-268) | Pass Target: >= 90% AviationExam

### 1. STRAPDOWN IRS ARCHITECTURE VS GIMBALLED INS

Inertial systems calculate present position, velocity, and attitude completely autonomously without receiving external radio signals or satellites:

- Gimballed INS (Older Technology): Mounted on a stabilized physical gimbal platform kept level with local Earth horizontal by torque motors. Heavy, mechanically complex, prone to gimbal lock during aerobatic maneuvers.
- Strapdown IRS (Modern Commercial Standard): Three Ring Laser Gyros (RLGs) and three solid-state accelerometers are rigidly 'strapped down' directly to the aircraft fuselage frame. A high-speed digital navigation computer mathematically converts body-axis accelerations into Earth-referenced horizontal navigation coordinates.

### 2. RING LASER GYRO (RLG) & THE SAGNAC EFFECT

Ring Laser Gyros have NO MOVING MECHANICAL ROTORS. Rotation is sensed using laser light optics via the Sagnac Effect:

- Operating Principle: Two monochromatic helium-neon laser beams travel in opposite directions (clockwise and counter-clockwise) around a sealed triangular cavity of Zerodur glass-ceramic via dielectric mirrors.
- The Sagnac Effect: When the aircraft rotates about the gyro axis, the path length for the beam traveling in the direction of rotation INCREASES, while the path length for the counter-rotating beam DECREASES. This path difference creates an optical frequency shift:  $\Delta f = (4 \times \text{Area} \times \Omega) / (\text{Wavelength} \times \text{Optical Perimeter})$ .
- Interference Fringe Detection: The two beams combine at a photodetector, creating optical interference fringes. The fringe transit rate is directly proportional to angular rotation rate.
- Laser Lock-In & Piezo-Dither Motor: At very low rotation rates ( $< 0.1^\circ/\text{sec}$ ), backscattering causes the two laser frequencies to lock together, producing ZERO output (dead band). To prevent lock-in, a piezo-electric dither motor vibrates the RLG back and forth mechanically at ~400 Hz through the lock-in zone!

### 3. IRS ALIGNMENT PHASE & SCHULER TUNING

| Phase / Principle                     | Operational Process & Duration                                                                                                                                                | Flight Crew Action & Mathematical Feature                                                                                                                              |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase / Principle</b>              | Operational Process & Duration                                                                                                                                                | Flight Crew Action & Mathematical Feature                                                                                                                              |
| <b>Initial Alignment (Stationary)</b> | Takes 5 to 10 minutes (longer at high latitudes, e.g. 15 min at 70°N).<br>AIRCRAFT MUST REMAIN COMPLETELY STATIONARY! (Cargo loading or wind buffeting will abort alignment). | Crew inputs present gate coordinates into MCDU. IRS measures Earth's rotation (15°/hr) to find TRUE NORTH (gyrocompassing) and senses gravity to level accelerometers. |
| <b>Latitude Limitations</b>           | IRS cannot align above Latitude 78° or 82° North/South.                                                                                                                       | At extreme polar latitudes, the horizontal component of Earth rotation is too weak for gyrocompassing.                                                                 |

**Schuler Tuning (84.4 Minutes)**

Accelerometers are mathematically tuned to oscillate like an 84.4-minute pendulum whose length equals Earth radius ( $R = 6,371 \text{ km}$ ).

Prevents aircraft acceleration from being mistaken for a tilt of the vertical. Schuler tuning contains navigation position errors within a periodic 84.4-minute bounded oscillation cycle!

**[!] EXAM TRAP (AVIATIONEXAM) - IRS Alignment Rules & Lat/Long Verification**

- The aircraft MUST NOT BE MOVED during the alignment cycle! - The flight crew must enter present position coordinates into the MCDU. The IRS compares entered latitude with its own internally sensed gyrocompass latitude: if discrepancy  $> 1^\circ$ , alignment FAILS! - Note: IRS navigates referenced to TRUE NORTH, not Magnetic North!

**STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)****[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): IRS Alignment & Sagnac Effect**

Step 1: Aircraft must be 100% stationary on ramp. Step 2: RLG sensors detect Earth rotation ( $15 \text{ deg/h} \times \cos(\text{latitude})$ ) to find True North automatically. Step 3: Accelerometers sense gravity vector to level the platform. Step 4: If entered gate position differs by  $>1$  degree latitude, ALIGN annunciator flashes.

**4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)****Q1: What is the physical optical phenomenon utilized by a Ring Laser Gyro (RLG) to measure angular rotation**

- [A]: Doppler effect
- [B]: The Sagnac effect, where optical path lengths differ between counter-propagating laser beams
- [C]: Photo-electric emission
- [D]: Faraday induction

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. The Sagnac effect dictates that rotating the laser cavity creates a path length difference between the clockwise and counter-clockwise beams, producing a frequency difference proportional to rotation rate.

**Q2: What is the purpose of the piezo-electric 'dither motor' installed on a Ring Laser Gyro?**

- [A]: To cool the laser mirrors
- [B]: To vibrate the gyro back and forth through the low-speed 'lock-in' threshold, preventing zero-output error
- [C]: To level the platform
- [D]: To align with magnetic north

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. At low rotation rates, internal mirror backscattering locks the two laser frequencies together (laser lock-in). A dither motor mechanically oscillates the gyro at high frequency to keep it out of the dead band.

**Q3: What is the period of the Schuler oscillation cycle in an inertial reference navigation system?**

- [A]: 24.0 hours
- [B]: 84.4 minutes
- [C]: 12.5 minutes
- [D]: 360 seconds

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. The Schuler pendulum period is  $T = 2 \times \pi \times \sqrt{R / g} = 84.4$  minutes.

Tuning the IRS to 84.4 minutes isolates linear accelerations from gravity vectors, preventing unbounded position drift.